

CSDS 302 Discrete Math | Fall 2026 | Homework 2

Instructions

Section 2.1: 2, 5bc(4 points each), 8be(4 points each), 10be(4 points each), 15, 19, 20, 22, 24, 29, 35, 43, 49, 51

Section 2.2: 2, 8, 14b, 16, 20be(4 points each), 22be(4 points each), 23be(4 points each), 27, 30, 41, 46cd(4 points each)

Section 2.3: 4, 9, 23, 29, 32, 40, 44

Section 3.1: 12, 17b, 20, 22a, 28

Section 3.2: 1, 5, 12, 17, 29, 48

Section 3.3: 2, 10ab (4 points each), 19, 23

Section 3.4: 2, 14, 22, 24

—

Feedback

Score: 87.25/100

Section 2.1

5c: -4, it is a statement as it has a definite truth value(false)

8b: -4, should be

$\sim w \wedge (h \wedge s)$

35: -4, $-1 < x \leq 1$

Section 2.2

8: -4, wrong truth table

16: -4, not logically equivalent

20e: -4, this negation shouldn't be in "if-then" form

22a: -0, correct answer but to the wrong question

28: -4

Section 3.2

-8

Section 3.3

2c: -4, false

23: -4

Section 3.4

-8

Please make sure you are answering the right questions

DISCRETE MATH (SSDS 302/MATH 304) HOMEWORK #2
ZELAN EROZ ESPANTO (zre3)

SECTION 2.1

- ② If all prime numbers are odd, then 2 is odd. However, since 2 is an exact factor of 2, it's not odd. Therefore, not all prime numbers are necessarily odd.
- ⑤ b. "she is a mathematics major" is a statement.
c. " $128 = 2^0$ " is not a statement.

- ⑧ $\sim \vee \wedge$: let h = "John is healthy" w = "John is wealthy"

$\underline{b}$ s = "John is wise"

$\sim h$ John is not wealthy

$w \wedge s$ John is wealthy & wise

$(\sim h \vee w) \wedge s$

$\underline{e}$ ~~not~~ "healthy & wise" $\rightarrow h \wedge s$

not (healthy & wise) $\rightarrow \sim(h \wedge s)$

wealthy but/and not healthy & wise $\rightarrow w \wedge \sim(h \wedge s)$

$\wedge$ but/and

$\vee$ or

$\sim$ not

- ⑩ $\underline{a}$ p = "DATAENDFLAG is off" q = "ERROR equals 0"
 r = "SUM is less than 1000".

$\underline{b}$ $p \wedge \sim q$ $\underline{c}$ $\sim p \vee (q \wedge r)$ or $\bar{p} \vee (q \wedge r)$

- ⑮ $p \wedge (\sim q \vee r)$

p	q	r	$\sim q$	$\sim q \vee r$	$p \wedge (\sim q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	F	T	F
F	T	F	F	F	F
F	F	T	T	T	F
F	F	F	T	F	F

2

(19)

p	t	$p \wedge t$	$p \wedge \neg t$ and p
F	F	F	F
F	T	F	F
T	F	F	F
T	T	T	T

They are logically equivalent

(20)

$p \wedge c$ and $p \vee c$	
$p \wedge c$	$p \vee c$
F F	F
F T	T
T F	T
T T	T

NOT LOGICALLY EQUIVALENT

(22)

22

$p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$

p	q	r	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
-----	-----	-----	-----------------------	----------------------------------

LOGICALLY EQUIVALENT

$\wedge$ And

$\vee$ Or

F	F	F	F	F
F	F	T	F	F
F	T	F	F	F
F	T	T	F	F
T	F	F	F	F
T	F	T	T	T
T	T	F	T	T
T	T	T	T	T

LOGICALLY EQUIVALENT

(24)

$p \vee q$	$(p \vee q) \vee (p \wedge r)$	$(p \vee q) \wedge r$
F F F	F	F
F F T	F	F
F T F	T T	F
F T T	T T	T
T F F	T T	F
T F T	T T	T
T T F	T T	T
T T T	T T	T

NOT logically equivalent

4x lat pull down
3-4 x row neck
6 sleep curls

(29) The computer program has logical error in the first ten lines or it is not being run with an incomplete data set.

(35) The negation of $x \leq -1$ or $x > 1$ is $x \geq -1$ and $x < 1$

(43) Tautology or contradiction

$\sim p \vee q$	p	q	$\sim p \vee q$	$p \wedge \sim q$	$(\sim p \vee q) \vee (p \wedge \sim q)$
F	F	T	T	F	T
F	T	T	T	F	T
T	F	F	T	T	T
T	T	T	T	F	T

It has all TRUE values $\therefore$ It's a tautology

$$\begin{aligned}
 (49) \quad & (p \vee \sim q) \wedge (\sim p \vee \sim q) \\
 (a) \quad & \equiv (\sim q \vee p) \wedge (\sim q \vee \sim p) \\
 (b) \quad & \equiv \sim q \vee (p \wedge \sim p) \\
 (c) \quad & \sim q \vee c \\
 (d) \quad & \sim q \\
 \therefore \quad & (p \vee \sim q) \wedge (\sim p \wedge \sim q) \equiv \sim q
 \end{aligned}$$

By commutative property
 ASSOCIATIVE PROPERTY
 BY NEGATION
 BY IDENTITY $q \vee c \equiv q$

(51) $p \wedge (\sim q \vee p) \equiv p$

SECTION 2.2

2, 8, 14b, 16, 20be(4 points each), 22be(4 points each), 23be(4 points each), 27, 30, 41, 46cd(4 points each)

2. “I am on time for work if I catch the 8:05 bus.” can be written as “If I catch the 8:05 bus, then I am on time for work.”

8.

p	q	r	$\neg p$	$q \rightarrow r$	$\neg p \vee (q \rightarrow r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

14b. If n is prime, then n is odd or If n is prime, then n is 2.

16

Consider

p : You paid full price.

q : You didn't buy it at Crown Books.

The symbolic form of “If you paid full price, you didn't buy it at Crown Books” is $p \rightarrow q$.

And the symbolic form of “You didn't buy it at Crown Books or you paid full price.” is

$\neg p \vee q$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The truth table of $p \rightarrow q$ and $\neg p \vee q$ are the same. So the two statements are logically equivalent.

20

b: “Today is New Year’s Eve and tomorrow is not January”

e: “If x is nonnegative, then x is not positive nor x is 0.”

22b: If tomorrow is not January, then today is not New Year’s Eve

22e: If x is negative, then x is not nonnegative. OR. If x is negative, then x is not positive and x is not 0.

23b

- Converse: If tomorrow is January, then today is New Year’s Eve.
- Inverse: If today is not New Year’s Eve, then tomorrow is not January.

23e

- Converse: If x is positive or x is 0, then x is nonnegative.
- Inverse: If x is negative, then x is neither positive nor 0.

27

A **conditional statement** is:

$$p \rightarrow q$$

The **converse** of the conditional statement is:

$$q \rightarrow p$$

The **inverse** of the conditional statement is:

$$\neg p \rightarrow \neg q$$

Truth Table

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

The above truth table shows that $q \rightarrow p$ and $\neg p \rightarrow \neg q$ are **logically equivalent**.

Logical Equivalence

$$q \rightarrow p \equiv \neg p \rightarrow \neg q$$

30

The truth table for $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ is constructed as follows.

p	q	r	$p \wedge q$	$p \wedge r$	$q \wedge r$	$p \wedge (q \wedge r)$	$(p \wedge q) \vee (p \wedge r)$	$p \wedge (q \wedge r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T	T
T	F	T	F	T	T	T	T	T
T	F	F	F	F	F	F	F	T
F	T	T	F	F	T	F	F	T
F	T	F	F	F	T	F	F	T
F	F	T	F	F	T	F	F	T
F	F	F	F	F	F	F	F	T

Since the last column is all T, $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ is a tautology.

Since the last column is all T, the statement $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ is a tautology.

41: If a triangle has two 45° angles, then it is a right triangle.

46

Let P = Compound X is boiling

Let Q = Its temperature must be at least 150Celsius.

C

(c) Compound X will boil only if its temperature is at least 150°C.

True

- P only if Q

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

- The last two columns are identical, thus the given statement is **true**.

d

(d) If compound X is not boiling, then its temperature is less than 150°C.

Not necessarily true

- If not P, then not Q

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$\neg P \rightarrow \neg Q$
T	T	F	F	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

- In the third row, $P \rightarrow Q$ is true, while $\neg P \rightarrow \neg Q$ is false, thus the given statement is **not necessarily true**.

Section 2.3:

4

“If this graph can be colored with 3 colors, then it can be colored with 4 colors”: $P \rightarrow Q$

“This graph cannot be colored with 4 colors”: $\neg Q$

Modus Tollens: Since we know that $P \rightarrow Q$ and we are given $\neg Q$, we can conclude that $\neg P$. Therefore this graph cannot be colored with three colors.

9

p	q	r	$p \wedge q$	$p \wedge q \rightarrow \neg r$	$p \vee \neg q$	$\neg q \rightarrow p$	Conclusion ($\neg r$)	Premises Hold	Valid Inference
True	True	True	True	True	True	True	False	True	False
True	True	False	True	True	True	True	True	True	True
True	False	True	False	False	True	True	False	False	False
True	False	False	False	True	True	True	True	True	True
False	True	True	False	False	False	True	False	False	False

Interpretation

- The “**Valid Inference**” column checks whether the premises hold and whether the conclusion ($\neg r$) is true.
- If the premises are **true** but the conclusion is **false**, then the argument is **invalid** in that case.
- Since there are cases where “Valid Inference” is **False**, the conclusion $\neg r$ is **not universally valid** based on the given premises.

23: The argument is valid if whenever all premises are true. The conclusion must also be true. Since there are cases where the premises hold but the conclusion does not, the argument is not valid. Solution is shown below.

Let M=“Oleg is a Math major”, E=“Oleg is an economics major”, C=“Oleg is required to take Math 362”

Premises:

1. MVE
2. $M \rightarrow C$

3. $E \vee \neg C$

M	E	C	$M \vee E$	$M \rightarrow C$	$E \vee \neg C$	Premises Hold	Conclusion ($E \vee \neg C$)	Valid Inference
T	T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	T	F
T	F	T	T	T	T	T	T	T
T	F	F	T	F	F	F	F	F
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T	T
F	F	T	F	T	F	F	F	F
F	F	F	F	T	T	F	T	F

Objective

The objective is to determine whether the argument is **valid** or **invalid** using a truth table.

Step 1: Representing the Argument in Symbolic Notation

Let:

- **M** = Oleg is a Math major.
- **E** = Oleg is an Economics major.
- **C** = Oleg must take Math 362 course.

Thus, the argument can be rewritten in symbolic form as:

[
 $M \vee E$
]

[
 $M \rightarrow C$
]

[
 $E \vee \neg C$
]

Step 2: Constructing the Truth Table

We construct the truth table by listing all **eight possible combinations** of truth values for **M, E, and C**.

M	E	C
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Step 3: Filling in Additional Columns

Using the definition of negation, we determine the values for $\neg C$:

M	E	C	$\neg C$
T	T	T	F
T	T	F	T
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Step 4: Computing Premises and Conclusion

Using the definitions of logical **OR** ($\vee$) and **Implication** ($\rightarrow$):

M	E	C	$\neg C$	$M \vee E$	$M \rightarrow C$	$E \vee \neg C$
T	T	T	F	T	T	T
T	T	F	T	T	F	T
T	F	T	F	T	T	F
T	F	F	T	T	F	T
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	F	T	F	F	T	F
F	F	F	T	F	T	T

Step 5: Checking for Validity

From the table, observe that **rows 1, 3, 5, and 6** are critical, as they make all **premises true**.

To check for **validity**, we verify if whenever **all premises are true**, the **conclusion is also true**.

- **However, row 3 reports a False conclusion**, meaning the argument is **invalid**.

Final Conclusion:

The argument is **Invalid**.

29 There is an inverse error.

32)

Let:

- **P** = I get a Christmas bonus.
- **Q** = I sell my motorcycle.
- **R** = I'll buy a stereo.

The argument to be proven **valid** is:

$$P \rightarrow R$$

Proof by division into cases:

$$Q \rightarrow R$$

Thus, we conclude:

$$P \vee Q \rightarrow R$$

Truth Table:

P	Q	R	$P \rightarrow R$	$Q \rightarrow R$	$P \vee Q$	$P \vee Q \rightarrow R$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	T	T	T	T
T	F	F	F	T	T	F
F	T	T	T	T	T	T
F	T	F	T	F	T	F
F	F	T	T	T	F	T
F	F	F	T	T	F	T

Conclusion:

The truth table verifies the logical validity of the statement $(P \vee Q \rightarrow R)$.

Let me know if you need any modifications!

Out of all four, one person speaks the truth. (*Others are lying.*)

Statements:

- **Lefty:** "Muscles was shooting craps with Socko when Sharky was knocked off."
(*Implies neither Muscles nor Socko killed Sharky.*)
- **Fats:** "Muscles is not a killer."
- **Socko:** "Lefty is the killer."
- **Muscles:** "Lefty is not the killer."

Analysis:

- Since **Lefty** and **Fats** both say that Muscles didn't kill Sharky, but both **cannot** be correct at the same time, one of them must be lying.
- Also, note that if **Socko lies**, then **Muscles speaks the truth**, or vice versa.
- Thus, **either Muscles or Socko speaks the truth**.
- So, **Lefty and Fats must be lying** (because only one person can be telling the truth).

Since **Fats is lying** (and Lefty also), then **Muscles must be the killer**.

Conclusion:

Thus, **Muscles is telling the truth** and **Muscles killed Sharky**.

44

Section 3.1

12: The statement is true. A geometric figure can be rectangle and square at the same time because a rectangle whose length and width are equal is a square.

17b: $4(n^2 + n + 1) - 3n^2$

$= 4n^2 + 4n + 4 - 3n^2$

$$= n^2 + 4n + 4$$

$$= (n + 2)^2$$

Hence $4(n^2 + n + 1) - 3n^2 = (n + 2)^2$. So the given statement is true.

20: The contrapositive of the given statement is: "There exists an x such that x has property p but has not property g"

22a

$\forall$ computers c , c has a CPU.

Let

D = All computers (domain)

$Q(x)$ = " c has a CPU"

We can then rewrite the given statement as:

$$\forall x \in D, Q(x)$$

The negation of the universal statement is:

$$\sim (\forall x \in D, Q(x)) \equiv \exists x \in D \text{ such that } \sim Q(x)$$

Replacing " $\in D$ " by "computer", " $Q(x)$ " by " x has a CPU" and $\sim$ by "not", we then obtain:

Negation = $\exists$ computer c such that c does not have a CPU

28: The proposed negation is correct.

Section 3.2:

1

- Correct. Since it directly states the negation.
- Wrong. This is a different universal statement.
- Wrong. Irrelevant to the negation.
- Wrong. It contradicts the original but is not its logical negation.
- Correct. Since "Some" means "at least one" which matches the negation.
- Wrong. It's a different universal statement.

5a: If graduation is counted then the earned grade is C in this course.

5b. If a number is divisible by 8, then it's divisible by 4.

12 $\forall$ denotes for all

17: It's TRUE because $P(c,n)$ means c is capital of n and in given statement paris is capital of France, so it's True.

29: Statement is true because the argument is in regular order and in sequence and hence a valid argument.

48a: "Every odd integer can be written as the sum of 1 and twice of some integer"

48b Negative of given statement is: $\exists$ odd integer n , $\neg \exists$ an integer k such that $n = 2k + 1$.

Section 3.3:

2

- a. True
- b. False
- c. True
- d. True

10a: $\exists$ a person x such that $\forall$ people y , x does not trust y .

10b: FALSE

19 "For every real number, there exists a real number such that sum of these two is 0."
The negation of this statement is " $\forall x \in R, \exists y \in R$ such that $x + y \neq 0$ " or "For every real number, there exists at least one real number for which their sum is not zero."

23 $X=g$

Section 3.4

2

14: Preceding number of a perfect square of an integer n can be written as $n^2 - 1$ which can be further factorized as $n^2 - 1 = (n - 1)(n + 1)$ and of perfect square is greater than 4 then value of n will be greater than 2 then both $n - 1$ and $n + 1$ will be

two integers other than 1 and $n^2 - 1$ dividing $n^2 - 1$. So preceding number of a perfect square greater than 4 will not be a prime number.

22 TRUE.

Let n is an odd integer. This implies that n is not divisible by 2. Now the negative of n is $(-n)$. $-n = -1(n)$

(-1) is not divisible by 2 and n is also not divisible by 2. Therefore $(-n)$ is not divisible by 2. This implies that $(-n)$ is odd.

24. VALID.

If we multiply an even integer with any integer it will always be even as the even integer will have 2 as a factor and hence any number multiplied by 2 will always be even.